

① If A and B are events such that $P(A/B) = P(B/A) \neq 0$, then

a) $A \subset B$ but $A \neq B$

b) $A = B$

c) $A \cap B = \emptyset$

d) $P(A) = P(B)$

② Find $\frac{dy}{dx}$, if $(\cos x)^y = (\cos y)^x$

③ Find: $\int \cos^3 x \cdot e^{\log \sin x} dx$

④ Find the value of K if $\sin^{-1} \left[K \tan \left(2 \cos^{-1} \frac{\sqrt{3}}{2} \right) \right] = \pi/3$

⑤ Verify whether the function f defined by

$$f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases} \text{ is continuous}$$

at $x=0$ or not?

⑥ Check for differentiability of the function f defined by $f(x) = |x-5|$ at the point $x=5$.

⑦ Find the vector equation of the line passing through the point $(2, 3, -5)$ and making equal angles with the coordinate axes.

⑧ If $x = a \sin^3 \theta$, $y = b \cos^3 \theta$ then find

$$\frac{d^2 y}{dx^2} \text{ at } \theta = \frac{\pi}{4}.$$